

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte(Deemed to be University)

First Semester M.Tech. (CBS) (CBCS) Degree Examinations

January 2023

22CBS102 – CYBER FORENSICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **one full** question from each Unit.

		Marks	BT*	CO*	PO*
Unit-I					
1.	a) Explain the auditing and security architecture in cyber forensics.	8	L*2	1	1
	b) Illustrate the structure of a hard disk in computer system.	8	L2	1	1
	c) What is digital evidence? List the different sources of digital evidence.	4	L1	1	1
2.	a) Give the difference between FAT and NTFS filesystem.	8	L2	1	1
	b) What is Metadata? Explain the different types of Metadata	8	L2	1	1
	c) Describe HKCU (HKEY_CURRENT_USER) and HKLM (HKEY_LOCAL_MACHINE) windows registry.	4	L2	1	1
Unit-II					
3.	a) Explain the Filesystem Hierarchy Structure (FHS) in Linux OS	8	L2	2	1
	b) Explain the commands involved in collecting the volatile data.	8	L2	2	1
	c) Explain boot loader, kernel and init system.	4	L2	2	1
4.	a) Explain the different types of Linux log files.	6	L2	2	1
	b) Give the difference between EXT 2/3/4 file systems.	6	L2	2	1
	c) Explain the working and syntax of the following commands i) dmseg ii) fsck iii) stat iv) history	8	L2	2	1
Unit-III					
5.	a) What is database and Explain the components of the database?	8	L2	3	1
	b) Explain the steps involved in performing the forensics on database.	8	L2	3	1
	c) What is transaction log and where do we see those transaction logs?	4	L2	3	1
6.	a) Explain the steps involved in recovering the database if it is deleted.	8	L2	3	1
	b) What is database tampering? Explain the different types of digital evidence in database	8	L2	3	1
	c) List the number of log files that get generated in database.	4	L2	3	1
Unit-IV					
7.	a) Explain tcpdump and fwmonitor tools in capturing and analyzing the packets.	10	L2	4	1
	b) Explain the following in wireshark i) what is wireshark? ii) what is promiscuous mode in wireshark? iii) List the pros and cons of wireshark. iv) List the usescases of wireshark.	10	L2	4	1
8.	a) Explain the functionalities of every layer in OSI model.	10	L2	4	1
	b) Explain how to apply the capture filter on protocols and ports.	6	L2	4	1
	c) List the command lines to filter on MAC address.	4	L1	4	1
Unit-V					
9.	a) Explain any use case pertaining to identity theft.	10	L2	5	1
	b) Explain the Indian perspective of cyber crime in framing the IT laws.	10	L2	5	1
10.	a) what is DOS attack? Explain the different types or levels of DOS attack and also list the tools used to launch the DOS attack.	10	L2	5	1
	b) List and explain the applications of digital forensics.	10	L2	5	1